



Islamic University of Technology

Lab Report

Experiment No: 02

Student No: 210041105

Course Code: PHY-4242

Name of the Experiment: STUDY OF OHM'S LAW

Date of Performance: 31/01/23

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Submitted by:

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Section: 01 (A)

Dept. of Computer Science and Technology (CSE)

Objective:

The purpose of this experiment is to introduce Ohm's law to the students. After completion of this experiment, the students will be able to understand the importance of Ohm's law in electrical circuits.

Theory:

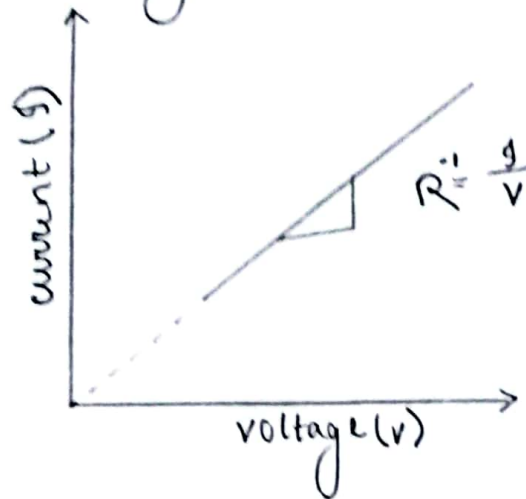
Ohm's law states that, "At any constant temp, the current flowing through any conductor is directly proportional to the voltage between the two ends of it."

If we represent voltage by V and current by I , then according to Ohm's law, we can write:

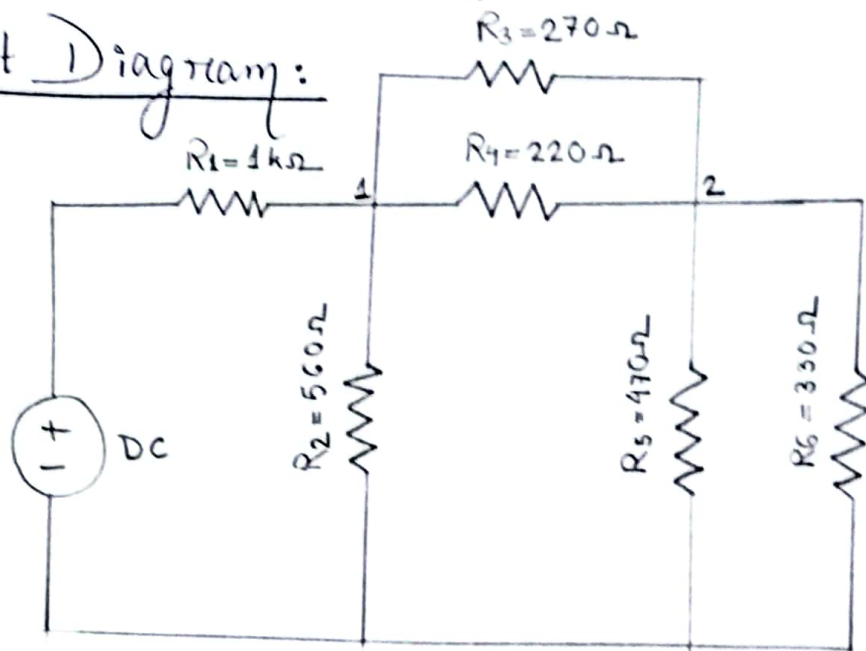
$$\begin{aligned} I &\propto V \\ \Rightarrow I &= V/R \\ \Rightarrow I &= \frac{1}{R} \cdot V \\ \Rightarrow I &= G \cdot V \end{aligned}$$

Here, $\frac{1}{R}$ is a constant which is known as conductance and R signifies the resistance

A graph drawn taking voltage and their corresponding currents would be linear in nature and would look something like this:



Circuit Diagram:



Data Collection:

Given power rating for each resistor =
The individual resistances after measuring,

$$R_1 = 992 \Omega$$

$$R_2 = 551 \Omega$$

$$R_3 = 205 \Omega$$

$$R_4 = 216 \Omega$$

$$R_5 = 454 \Omega$$

$$R_6 = 324.8 \Omega$$

Data Calculation:

$$P_{\max} = 0.25 \text{ W}$$

$$\Rightarrow I_{\max}^2 \times R_1 = 0.25 \text{ W}$$

$$\Rightarrow I_{\max} = \sqrt{\frac{0.25}{992}}$$
$$= 15.87 \text{ mA}$$

$$\therefore V_{\max} = I_{\max} \times R_1$$
$$= 15.87 \times 992$$
$$= 15.7480 \text{ V}$$

Supply Voltage (V)	Voltage Across R_C (V)	Current through R_C (mA)
0	0.037	0
2	1.954	1
4	4.12	1.9
6	6.14	2.5
8	8.13	2.8
10	10.04	3.5
12	12.04	4.2
14	14.00	4.9

Data Analysis:

A straight line is observed by plotting the current vs. voltage graph for R_C because the equation was linear.

The slope of the line represents conductance, $G = \frac{I}{V} = \frac{1}{R}$, which is the inverse of resistance.

From the graph,

$$\text{slope} = \frac{(4.2 - 3.5) \times 10^{-3}}{12.04 - 10.04} = 3.5 \times 10^{-4}$$

$$\therefore R = \frac{1}{\text{slope}} = 2857.143 \, \Omega$$

Discussion:

We obtain a linear curve plotting Voltage and Current accordingly. The discrepancy occurs due to the recording of source voltage.

However, if the voltage was taken along two ends of R_s , we would have obtained an accurate straight line. The slope ultimately represents conductance. Conductance means the measure of how easily electricity flows along a certain path through an electrical element. The inverse of conductance is resistance and resistance means the property of obstruction of current flow. The unit of conductance and resistance are Siemens and Ohm respectively.

We can conclude:

$$R = \frac{1}{G}$$

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

COURSE : EEE 4242 (ELECTRICAL AND ELECTRONIC FUNDAMENTALS LAB.)

EXPERIMENT NO. 2

NAME OF THE EXPERIMENT : STUDY OF OHM'S LAW

OBJECTIVE:

The purpose of this experiment is to introduce **Ohm's law** to the students. After completion of this experiment the students will be able to understand the importance of **Ohm's** in electrical circuits.

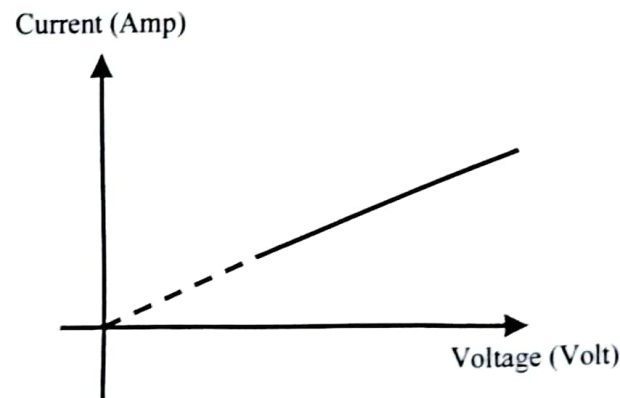
Ohm's Law: *At any constant temperature the current through any conductor is directly proportional to the voltage across it.*

If the voltage is represented by V and current by I then according to **Ohm's law**:

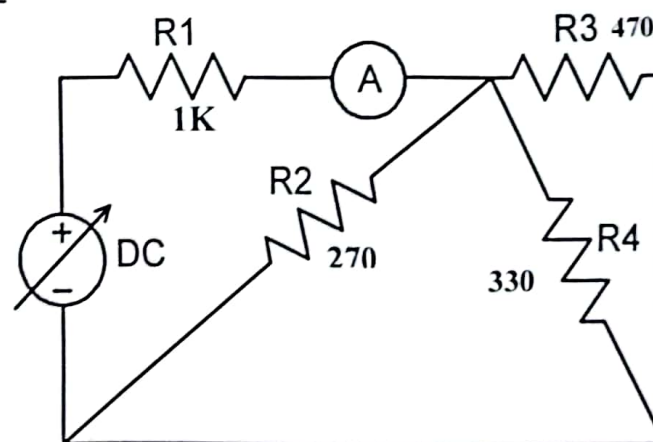
$$I \propto V$$

$$\text{or, } I = V/R$$

where $1/R$ is a constant and R is the resistance of the conductor.



Circuit Diagram:



Laboratory Tasks:

- Calculate maximum allowable DC voltage for the above circuit. The power rating of each resistor is 0.25Watt.
- Measure individual resistances and record their values. Construct the circuit.
- Measure current through R1 resistor and voltage across the same for different values of supply voltage up to maximum allowable voltage calculated before (increment by 2V at each step from 0V).
- Submit completed data sheet to your teacher for verification.

Sample Data Sheet

Supply Voltage	Voltage across R1	Current Through R1
0 V	0	0
2 V	.	.
4 V	.	.
6 V	.	.
8 V	.	.
10 V	.	.
.	.	.

Report

- i. Plot *Current* vs. *Voltage* and explain the shape. The plot should be straight line and if extrapolated it will pass through origin. What does the slope of the line represent?
- ii. Define *conductance* and *resistance*.
- iii. Calculate resistance from the curve and compare with measured resistance value. Show calculations in your report.
- iv. Discuss on your result.

